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Experimental study of the sound emitted by water drops impacting on a water surface

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Abstract An undergraduate-level experiment is described for studying the sound emitted as a liquid drop falls into a water bath. The falling drop intercepts a light beam which, after a preset delay, triggers a flash gun and an oscilloscope. This allows simultaneous photographs of the impact and oscilloscope traces of the sounds picked up by a microphone and hydrophone to be obtained. Such studies show that most of the sound emission originates from radial oscillations of an air-filled bubble lying within the bulk of the liquid and created by the impacting drop. The observed dependence of emission frequency on bubble radius agrees with that predicted by Minnaert's formula. In addition a small low-frequency signal is recorded as the drop strikes the surface, before the detachment of the bubble. Most of this latter signal is thought to be a near-field effect.

Zusammenfassung Ein Studentenversuch wird beschrieben, der den Schall untersucht, den ein Flüssigkeitstropfen verursacht, der in ein Wasserbad fällt. Der fallende Tropfen unterbricht einen Lichtstrahl, der nach einer voreingestellten Verspätung ein Blitzgerät und ein Oszilloskop aktiviert. Diese Aufstellung macht es möglich, dass man gleichzeitige Photographieren des Aufpralls und der Oszilloskopspuren des Schalls, den ein Mikrophon und ein Hydrophon aufnehmen, erhält. Solche Studien zeigen, dass die meiste Schallaussendung ihren Ursprung in radialen Schwingungen einer luftgefüllten Blase innerhalb der Masse der Flüssigkeit hat, die von dem aufrallenden Tropfen erzeugt wurde. Die beobachtete Abhängigkeit der Emissionsfrequenz vom Radius der Blase stimmt mit der von Minnaert's Formel vorhergesagten, überein. Zusätzlich wird ein kleines Signal von niedriger Frequenz registriert, wenn der Tropfen, bevor die Blase sich ablöst, auf die Oberfläche auftrifft. Das meiste in diesem letzten Signal ist, meint man, eine Nahfeldwirkung.

1. Introduction

In his book *The World of Sound*, Sir William Bragg (1921) suggests that the sounds emitted when liquid drops strike a liquid surface originate from the cavities created in the surface of the liquid by the impacting drops. He cites the (previously unpublished) work of Sir Richard Paget who modelled these cavities (as photographed by Worthington (1897)) out of plasticine and found that by blowing across the opening in them sounds were produced which were similar to those heard during liquid-drop impact. This resonant cavity model had in fact been first put forward by Mallock (1919). An alternative

model was proposed by Minnaert (1933) who suggested that the sound emission originated mainly from radial oscillations of spherical gas bubbles entrapped in the liquid. Although Minnaert did not discuss how these bubbles might be formed, the photographic studies of Richardson (1948) and Franz (1959) have shown that the start of the sound pulse (or 'plop') produced by an impacting liquid drop coincided with the formation of a bubble in the liquid.

The purpose of this article is to describe an undergraduate-level experiment aimed at simultaneously recording the sounds produced during liquid-drop impact and photographing the impact. A parallel paper (Leighton and Walton 1987) discusses in more detail the nature of the sound emitted from

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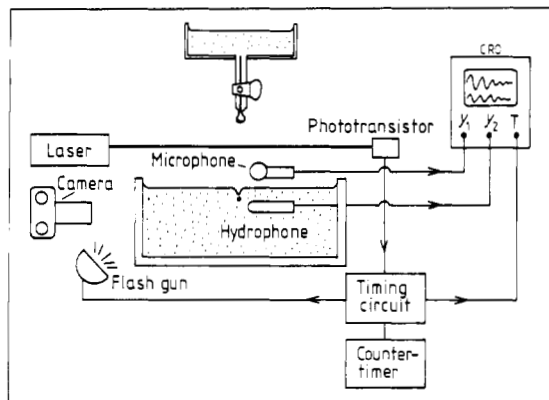


Figure 1 Experimental arrangement for recording the sound emitted by a liquid drop impacting on a liquid surface and for photographing the impact. T is the external trigger.

bubbles formed by blowing gas through nozzles located within the bulk of the liquid.

2. Experimental techniques

Figure 1 shows a schematic diagram of the equipment. Water from a reservoir (of sufficient cross-sectional area to ensure that the hydrostatic head remains effectively constant throughout an experimental run) is fed via a tap to a nozzle made by drawing out a thick-walled glass tube and squaring off the end on a grinding wheel. As the resulting drop descends it intercepts a laser beam, thereby cutting off the light which illuminates a phototransistor. This triggers a timing circuit which, after a variable preset delay, in turn triggers a photographic flash gun (Sunpak 24SR). By varying this time delay one can thus obtain a sequence of photographs showing the nature of the drop-liquid impact (admittedly for different, though identical, drops). The photographs were taken with a single-lens reflex camera (Nikon F2) fitted with a macro lens (plus extension tubes for the highest magnification work). The camera is set on 'bulb' and the shutter is

opened just prior to releasing the drop. The sounds emitted by the impacting drop were picked up in air by a dynamic microphone (frequency response 50 Hz to 14 kHz ± 3 dB) and under the water by a hydrophone (Brüel and Kjaer type 8103; frequency response 0.1 Hz to 200 kHz (-10 dB)). These sound signals were amplified and fed to the y_1 and y_2 inputs of a dual-beam storage oscilloscope which was triggered externally by a pulse from the timer circuit.

The time-delay circuit used to trigger the flash gun and oscilloscope is shown in figure 2. A decrease in the light intensity at the phototransistor (BDX 25) raises its resistance and produces the negative pulse required to trigger the timer circuit. This is based on a NE 555 chip in monostable mode; on receiving a negative pulse at the input, its output goes high for a time determined by the 100 k Ω variable resistor and the 1 μ F capacitor. This time is measured by an external counter-timer. The two 50 k Ω presets are used to set the quiescent input voltage to just above threshold. The logic circuit is designed to provide an output if the detector has produced a pulse and if the timer has waited its set time. It consists of three NOR gates in a CMOS 4001 logic chip. Gates 1 and 2 form a flip-flop; one output is always high, the other low. The inputs are usually both kept low; if one goes high its corresponding output goes low. To disarm the circuit the switch is set to B so that the output of gate 1 is high. To arm the circuit the switch is set to A but the output from gate 1 stays high until the timer starts, when the output from gate 2 goes high. Gate 3 now has one input high and one low. When the timer finishes its cycle its output drops so gate 3 now has both inputs low; its output goes high, triggering the 2N4174 thyristor—and hence the flash gun—and providing a triggering pulse to the oscilloscope.

3. Experimental procedures

Preliminary qualitative experiments suggested that many of the 'splashes' produced by an impacting drop were essentially silent and that those impacts

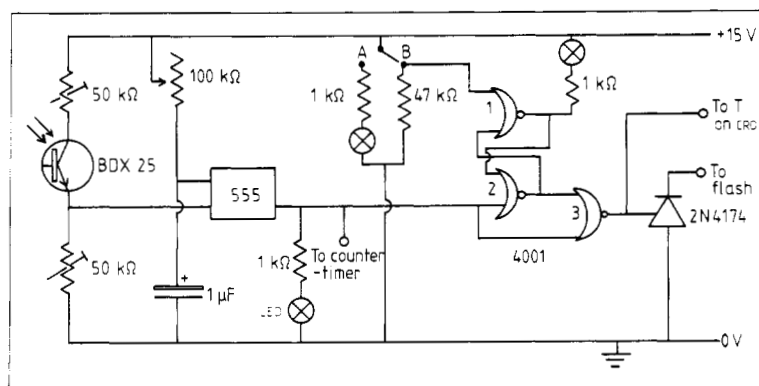


Figure 2 The timing circuit used to fire the flash gun and to trigger the oscilloscope in the equipment shown in figure 1.

which do give a metallic 'ping' also lead to the production of one or more bubbles in the liquid. A process of trial and error is required to determine the optimum height of fall of a drop from any given nozzle. Thus, for example, drops of volume 0.03 cm^3 (deduced by counting drops into a measuring cylinder) falling from a height of 7 to 8 cm give a fairly reproducible tone at around 7 kHz.

A quantitative study proceeds by setting the delay potentiometer to some arbitrary value and arming the circuit (by setting the switch of figure 2 at position A). As a falling drop intercepts the beam the timer is started and after the set interval (read on a counter-timer) the flash gun fires and, simultaneously, the oscilloscope is triggered. Figure 3(a) shows a representative photograph and figure 3(b) the associated microphone and hydrophone signals recorded on the oscilloscope. This demonstrates that the formation of a bubble coincides with the beginning of a damped simple harmonic oscillation. Owing to the time taken for the sound to travel to the microphone and hydrophone the photograph is actually taken at the times marked on the trace. This entire procedure is now repeated using a different delay time. By such means it is possible to build up a sequence of photographs showing the developing nature of the drop-liquid impact and the corresponding evolving sound signals. This is only possible because the sound signals produced are similar for all impacts in the sequence.

Figure 4 shows schematically the developing impact and the corresponding sound signals recorded by the hydrophone and the microphone as a function of time for 0.03 cm^3 drops falling from a height of 8.0 cm. Figure 5 shows the sequence of photographs used in compiling figure 4. Figure 4 shows that the hydrophone signal comprises an initial sharp rise coinciding with the arrival of the drop at the liquid surface, followed by a long oscillation which accompanies the development of a crater in the liquid surface. The pronounced damped simple harmonic signal beginning at around 27 ms coincides with the detachment of a bubble from the base of the now-rounded crater (see figure 3). Unlike the initial signal recorded in the 0–27 ms range, this harmonic oscillation is recorded by both hydrophone and microphone; the ratio of the two sound pressure amplitudes is roughly that expected for a sound signal passing from water to air. The initial signal recorded by the hydrophone alone is largely a near-field signal as was verified by increasing the distance between the hydrophone and the point of impact of the drop.

5. Discussion

5.1. The initial sound signal

The amplitude V_1 of the initial hydrophone signal (see figure 4) was measured as a function of the height h through which a drop fell. A graph of $\ln V_1$

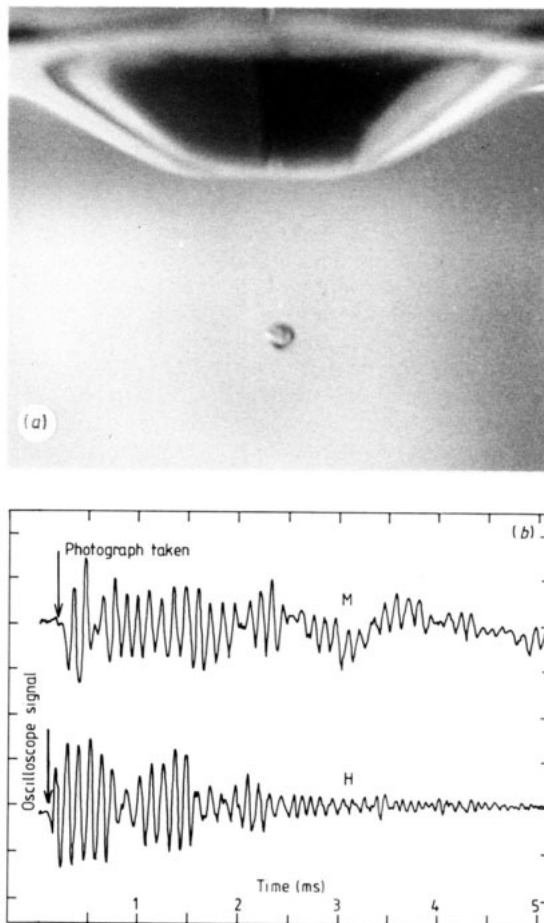


Figure 3 (a) Photograph of the water surface some 25 ms after impact of a drop of volume 0.03 cm^3 which had fallen from a height of 8.0 cm. (b) The corresponding microphone (M) and hydrophone (H) signals produced by this drop; the vertical arrows indicate the times at which the impact had the form shown in (a). Each division along the ordinate corresponds to 0.2 V for the microphone and to 0.5 V for the hydrophone (the signals are preamplified by different amounts before being fed to the oscilloscope).

plotted against $\ln h$ (figure 6) was found to be linear with a slope of 1.0 ± 0.1 both for drops of volume 0.03 cm^3 and for drops of volume 0.1 cm^3 . This, of course, implies that $V_1 \propto h$ or, since the speed of impact v of the drop with the liquid surface is given by $v^2 = 2gh$, that $V_1 \propto v^2$. Since the output voltage of the hydrophone is proportional to the acoustic pressure p in the liquid this becomes

$$p \propto v^2. \quad (1)$$

Once the liquid drop, impacting at speed v , fuses with the surface of the liquid the initial local speed of the liquid along a streamline will be v ; away from the surface it will drop to zero down the streamline.

Bernoulli's theorem now tells us that the pressure drop along the streamline will be proportional to v^2 . Although it is tempting to apply such an argument to explain the form of equation (1), this does not stand up to close scrutiny. The initial sharp rise in the hydrophone signal (see figure 4) probably originates from a 'water hammer' effect occurring before continuity of flow between the impacting drop and the underlying surface is established (Nystuen 1986). Because of the unreproducible nature of this part of the trace it was not possible to check out the expected linear dependence of its magnitude on v . The 'period' T_1 of the hydrodynamic pulse (see figure 4) was found to increase as a low power of h , typically around 0.2. The reason for this is uncertain.

decay in a roughly exponential manner. It is therefore reasonable to seek to explain this signal using the model first proposed by Minnaert (1933) in which the sound emission is supposed to originate from radial oscillations of entrapped spherical gas bubbles. As shown elsewhere (Leighton and Walton 1987) it is a simple matter to derive the radial resonant frequency ν of such a cavity by equating the maximum potential energy stored in the gas in the compressed bubble to the maximum kinetic energy of the surrounding liquid. The latter term is obtained by integrating the kinetic energy within a spherical shell of liquid of radii r and $r + dr$, namely $\frac{1}{2}(4\pi r^2 \rho dr) \dot{r}^2$ where ρ is the liquid density, from $r = a$ (the bubble radius) to $r = \infty$. This leads directly to the expression

$$\nu = \frac{1}{2\pi a} \left(\frac{2\kappa p_0}{\rho} \right)^{1/2} \quad (2)$$

5.2. The Minnaert oscillation

The experiments show that the pronounced damped simple harmonic oscillation picked up by both the microphone and the hydrophone starts at the instant of detachment of the bubble from the tip of the crater created by the impacting liquid drop (see figures 4 and 5). In some of the impacts the signals

where p_0 is the ambient hydrostatic pressure (to a good approximation the atmospheric pressure) and κ is the so called polytropic index which has a value of unity in an isothermal process and a value of γ in a reversible adiabatic process (in general, κ can be

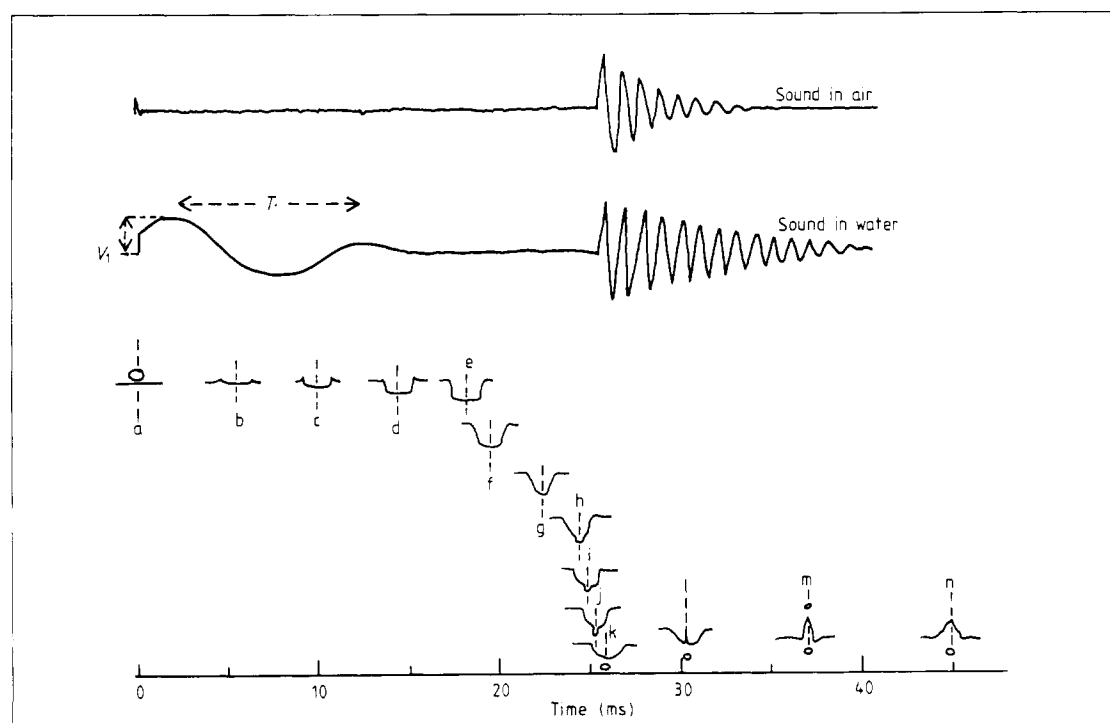


Figure 4 A schematic diagram showing how the form of the water surface and the sound signals emitted in air and in water vary with time. The drawings of the surface are displaced vertically for clarity. The envelope of the damped oscillation beginning at around 25 ms is usually more ragged than is suggested here (see, for example, figure 3(b)). The period of this oscillation is not shown to scale (in this experiment it has a value of around 7 kHz). These data refer to a drop of volume 0.03 cm^3 falling from a height of 8.0 cm.

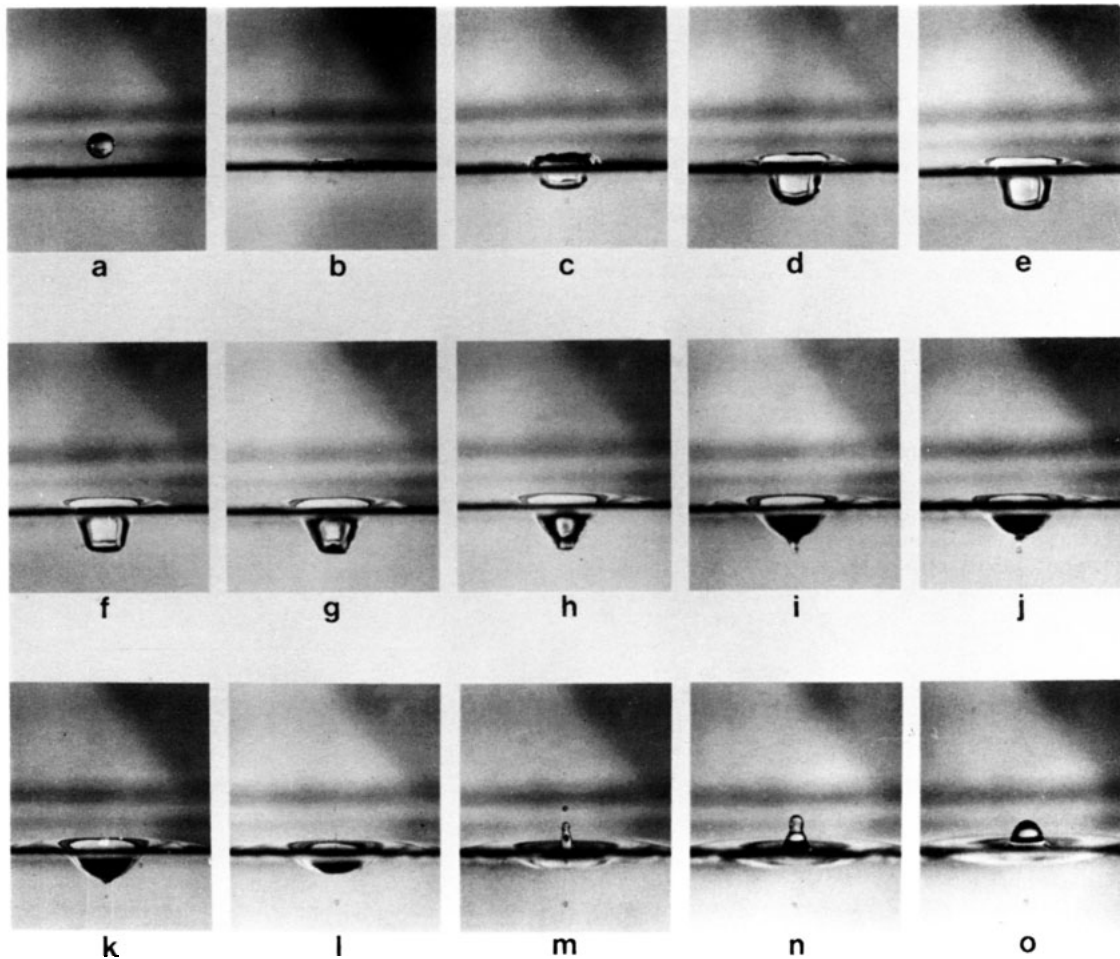


Figure 5 The sequence of photographs of a water drop impact as used in compiling figure 4. The time interval after impact at which each photograph was taken is given in figure 4.

expected to lie between these limits). To check out the correctness of this mechanism, measurements were made of the radius a of a detaching bubble and of the period ν of the emitted sound (as in figure 3) and the results plotted as a graph of a against ν^{-1} (see figure 7). Also shown in this graph is the theoretical dependence predicted by equation (2) for the cases $\kappa = 1$ and $\kappa = \gamma = 7/5$. This shows that, to within the experimental errors, the bulk of the data points lie within the expected limits. In practice most of the error lies in the measurement of a bubble diameter; the detaching bubble is often blurred (due probably to its rapid speed of detachment from the crater) and non-spherical (in which case a is taken as the geometric mean of the two extreme radii).

Although exponentially damped signals are sometimes observed, the traces more often resemble those shown in figure 3(b). Detailed study of these

envelopes (Pumphrey 1988) has shown that they arise from beating between normal modes of the tank, which are excited by the original signal.

In some of the traces the measured frequency of the damped oscillations decreases by some ten per cent over the duration of the signal. This may be attributed to the fact that the bubble is initially located at around some five bubble radii from the base of the crater (reformed after the bubble is ejected) and is finally located within the liquid bulk. Now the Minnaert relation assumes that the bubble lies within the bulk of the liquid since it integrates the kinetic energy term from $r = a$ to $r = \infty$. To estimate the effect of proximity to the free surface one might try integrating from $r = a$ to $r = 5a$. It is not difficult to show that such a change of limits accounts for the measured frequency drift. Strasberg (1953) gives an exact calculation of the Minnaert frequency of a bubble lying close to a surface.

It is possible to use the sequence of photographs of a developing cavity to check out Bragg's suggestion that the sound emission arises from standing waves set up in the cavity. If we consider a cavity such as that seen in the 15–25 ms time interval of figure 4 to be equivalent to an open-ended 'organ pipe' of length $\lambda/4$, where λ is the wavelength of the emitted sound, this does indeed predict roughly correct frequencies for the damped oscillatory signal when due allowance is made for end corrections to the pipe. However Bragg's model must be rejected as it fails to account for the absence of the oscillatory signal at earlier times (up to 25 ms in the case of figure 4).

Conclusions

In a recent hydrophone study of the underwater sound produced when various types of precipitation fall on a lake, Scrimger (1985) found that light rain produced a peak in the acoustic power spectrum at around 13 kHz. These measurements were made by feeding the hydrophone recordings directly into a spectrum analyser. Had such recordings been displayed on an oscilloscope it is possible that the 13 kHz peak might have been found to have been associated with characteristic bubble oscillations. If one is looking for underlying mechanisms it is therefore important to examine the signals on an oscilloscope before feeding them to a spectrum analyser.

Figure 6 Plots of $\ln V_1$ against $\ln h$ for drops of volume 0.03 cm^3 (A) and 0.1 cm^3 (B). Here V_1 is the initial peak in the hydrophone signal following the liquid drop impact (see figure 4) and h is the distance through which the drop has fallen.

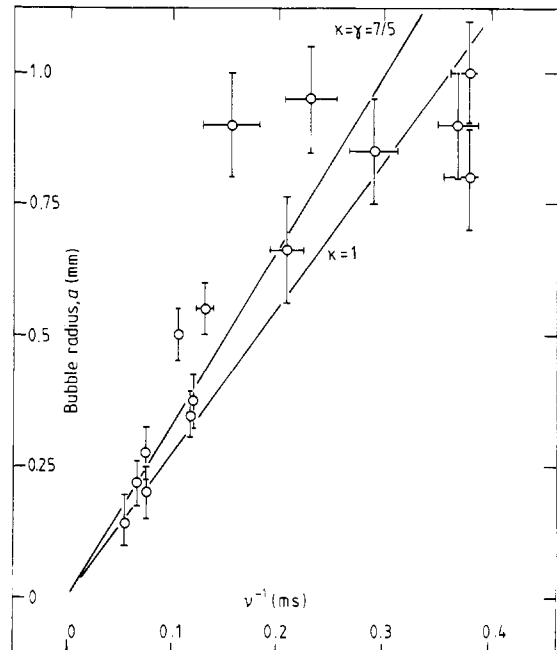
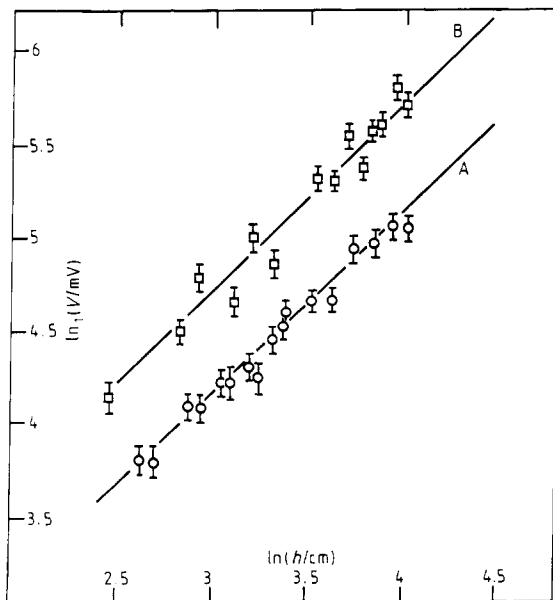


Figure 7 A plot of a against ν^{-1} , where a is the radius of the bubble and ν is the frequency of the emitted sound. These data refer to a variety of drops in the 0.03 to 0.1 cm^3 volume range dropped from heights between 8 and 30 cm .

Although, as these studies show, much of the sound emission following liquid-drop impact on a liquid surface can simply be accounted for by Minnaert oscillations of a gas-filled bubble, there is no such easy explanation of the initial low-intensity emission. Being largely confined to the liquid phase this initial emission in any event contributes little to the familiar 'plop' of a landing drop.

From a teaching perspective this project allows physics students to make measurements of a natural process about which comparatively little is known and to analyse their data in terms of familiar undergraduate concepts. It also allows them to assess the validity of two competing models of a phenomenon.

Acknowledgments

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